

SUKRITI TALWAR

24/CS/445

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EDUCATION

Year	Degree / Examination	Institute	GPA / %
2028	B.Tech in Computer Science & Engineering	Delhi Technological University, Delhi	9.279
2024	CBSE Class XII	Apeejay School, Pitampura, New Delhi	94.8%
2022	CBSE Class X	Apeejay School, Pitampura, New Delhi	96.0%

WORK EXPERIENCE

Indian Army, Directorate General of Information Systems | AI/ML Intern (June 2025 – August 2025)
Multimodal AI Pipeline for Defense Applications

- Developed **SAMPURNA**, a **multimodal AI pipeline** unifying **image segmentation (OpenCV)**, **speech processing & structured data fusion** for defense applications; automated **PyTorch** feature extraction, cutting preprocessing effort by **70%** across **3 modalities**.

ACADEMIC PROJECTS

SpeakEZ – Real-Time Speech Coaching Platform | *CNN/LSTM, MFCC, NLP, Multimodal AI* | [GitHub](#)

- Engineered parallel **audio & NLP pipelines** extracting **6+ speech signal dimensions** (pace, stress, fillers, posture) via **MFCC + CNN/LSTM** with **<200 ms** end-to-end latency.
- Deployed **live scenario-aware feedback** via **Multimodal AI** across **3 domains**; boosted pilot scores by **40%** with **edge-optimized** on-device inference module.

GESA – Fairness-Aware Allocation Engine | *GNNs, IntBERT, NSGA-II, Hugging Face* | [GitHub](#)

- Pioneered a fairness-aware matching engine to eliminate bias in large-scale allocation using **IntBERT embeddings**, **NexusGNN graph modeling & NSGA-II optimization**; processed **20,000+ profiles** and **3,000+ opportunities** via semantic matching.
- Delivered **94.5% Top-3 accuracy**, **0.98 fairness score**, and **37% diversity improvement** over baseline recommendation systems on benchmark evaluation splits.

Fashion Recommendation & Virtual Try-On System | *PyTorch, OpenCV, IDM-VTON, scikit-image* | [GitHub](#)

- Architected a **photorealistic garment try-on system** tackling cloth warping & texture preservation via **IDM-VTON diffusion model**; constructed **Image Encoder → Adapter → Denoising U-Net → VAE Decoder** pipeline with **OpenCV** geometric alignment.
- Optimized **scikit-image** preprocessing and **segmentation-mask** refinement for **real-time inference**; attained **87% visual similarity** on test garments, slashing **latency by 30%** over baseline.

PAG-CRAFT Thermal Super-Resolution Framework | *PyTorch, Computer Vision, Cross-Attention* | [GitHub](#)

- Innovated a **physics-aware super-resolution** framework to overcome **low-resolution thermal imaging** limitations using **optical-thermal cross-attention** & emissivity constraints in **PyTorch**; fused features across **2 sensor modalities**.
- Improved **PSNR by 2.1 dB** and **SSIM by 0.06**; reduced **reconstruction error by 18%** over object detection & segmentation baselines across **3 benchmark datasets**.

ACHIEVEMENTS

- Grand Finalist, Smart India Hackathon (SIH) 2025** – National-level Hackathon, Government of India.
- Winner, ArmorIQ Claw & Shield Hackathon** – BITS Pilani APOGEE 2026.
- Runner-Up & Best All-Girls Team, BrAInwave 2.0** (North India's Largest Hackathon) – SCAILE Track.
- Top 10 Finalist, ISRO Bhartiya Antriksh Hackathon 2025** – Space technology challenge by Indian Space Research Organisation.
- Runner-Up, Kriti Social Impact Challenge** – BITS Pilani TechFest 2026.
- 2nd Runner-Up, Product Case Competition 2.0** – BITS Pilani APOGEE 2025.
- Runner-Up, E-Conjecture** – Economics & Public Policy Case Competition, IIM Indore 2024.

SKILLS

Programming Languages: Python, C, C++, SQL, HTML, CSS

Data Analytics: Pandas, NumPy, Scikit-Learn, Matplotlib, Power BI, Excel

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, FastAPI, SciPy, Hugging Face

Technologies: Machine Learning, Data Analytics, Data Visualization, Predictive Modeling, Customer Segmentation, NLP, Computer Vision

Core Courses: Data Structures & Algorithms, DBMS, Operating Systems, OOPs, Probability & Statistics

Certifications: Intro to ML, Machine Learning Specialization, Deep Learning Specialization (Coursera – Andrew Ng)

POSITIONS OF RESPONSIBILITY

Co-Head, AI & ML Society (AIMS), DTU (August 2024 – Present)

- Mentored **90+ students** in **AI/ML, Computer Vision & Deep Learning**; guided teams to wins at BITS Pilani APOGEE 2026 and national-level competitions.
- Spearheaded **6+ hands-on bootcamps** (classical ML to LLMs), growing active membership by **40%** within one academic year.

Co-Head, ACM – DTU Student Chapter (September 2025 – Present)

- Orchestrated **5+ technical events** & coding contests, championing computing education among **500+ members** and driving chapter engagement up by **35%**.